Care of Critically Ill Patient

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Objectives

- adopt a structured, comprehensive approach to managing critical ill patients
- judge which patients are at most risk and plan to reduce their risk of adverse outcomes
- recognize the deteriorating patient and intervene to correct the problem
- acknowledge the importance of communication in managing critical ill patients

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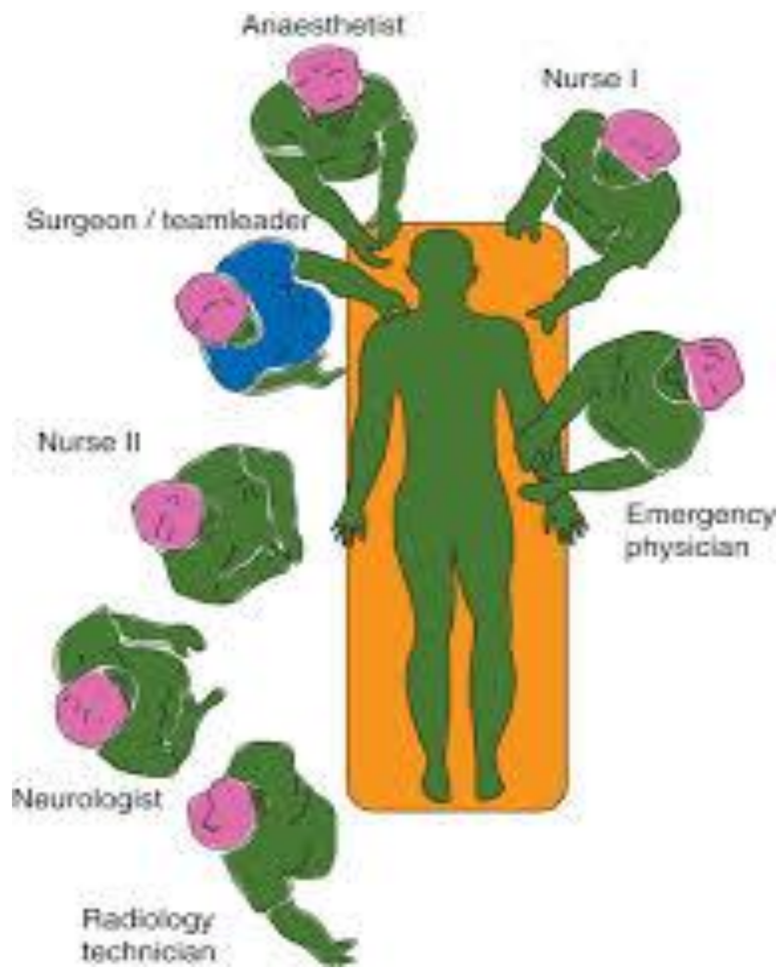
Definition

Critical Care Medicine has evolved as a dynamic, multidisciplinary field focused on the care of patients with life-threatening diseases

The Trauma Team

Trauma patient resuscitation is undertaken by a team of doctors and nurses; in this way, several tasks can be undertaken simultaneously



Immediate care
Advanced Trauma Life Support

- The initial management is considered in four phases:
- Primary survey
- Resuscitation
- Secondary survey
- Definitive care

The first two phases are undertaken simultaneously

The secondary survey, or head-to-toe examination of the patient, is started after the patient had been resuscitated adequately

Primary survey and resuscitation

- The primary survey ('ABC principles') comprises a sequential search for immediately life-threatening injuries:

A Airway with cervical spine control

B Breathing

C Circulation and hemorrhage control

D Disability: a rapid assessment of neurological function

E Exposure: while considering the environment and preventing hypothermia

Airway and the cervical spine The priority during resuscitation of any severely injured patient is to ensure a clear airway and maintain oxygenation

Use basic airway manoeuvres, with or without adjuncts such as an oropharyngeal and nasopharyngeal airway

Give the patient high-concentration O2 ; in the unintubated, spontaneously breathing patient, this is delivered with a mask and reservoir (non-rebreathing) bag

Assume the presence of a spinal injury in any patient who has sustained significant blunt trauma, until clearance procedures have been completed

This implies that the patient has been examined by an experienced clinician and radiological procedures have been completed

Reliable clinical examination cannot be obtained if the patient

- Has sustained a significant closed head injury
- Is intoxicated
- Has a reduced conscious level from any other cause
- Has significant pain from an injury, which 'distracts' attention from the neck

Tracheal intubation

-Indications for immediate intubation of the severely injured patient include:

- Airway obstruction unrelieved by basic airway manoeuvres
- Impending airway obstruction, e.g. from facial burns and inhalation injury
- GCS<9 (to facilitate CT scanning, intubation is required in many patients with a GCS>9)
- Haemorrhage from maxillofacial injuries compromising the airway
- Respiratory failure 2° to chest or neurological injury
- The need for resuscitative surgery
- Uncooperative patients requiring further investigations

The best technique for emergency intubation of a severely injured patient with a potential cervical spine injury is:

- Manual in-line stabilization of the cervical spine by an assistant whose hands grasp the mastoid processes and hold the head down firmly on to the trolley; this reduces neck movement during intubation. Do not apply traction to the neck

- Preoxygenation

Continue

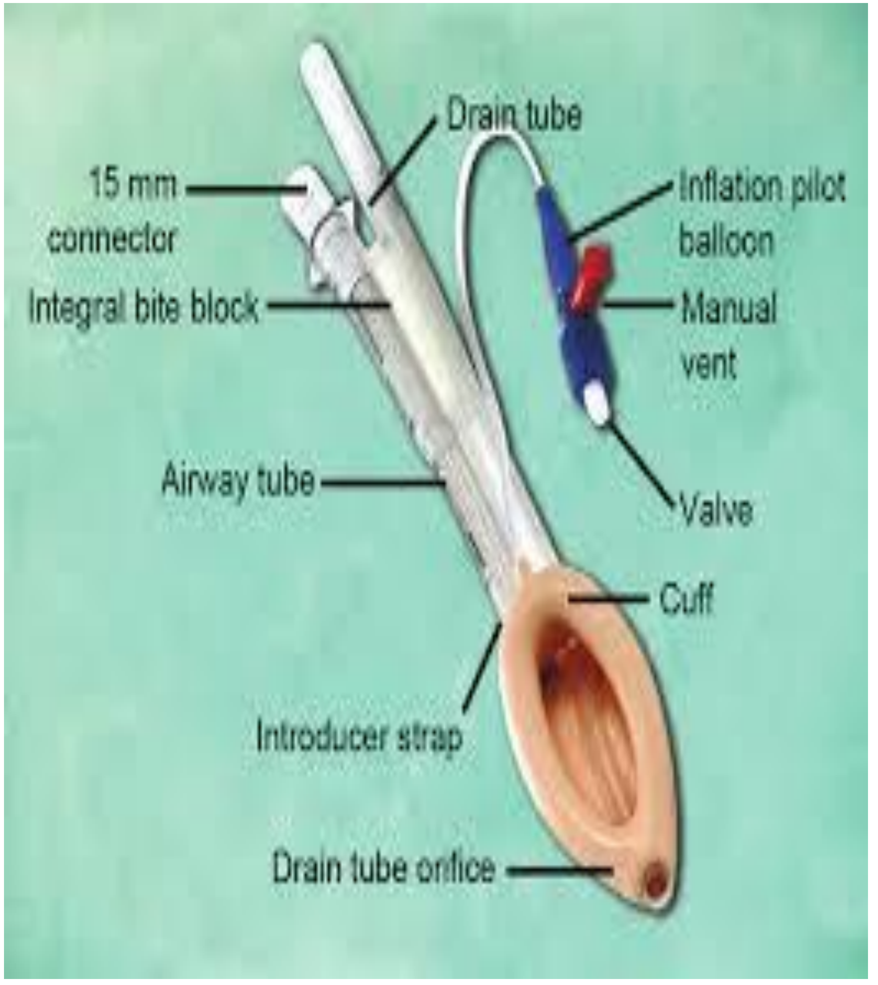
- IV induction of anesthesia. All induction drugs have the potential to produce or worsen hypotension, and the choice of drug is less important than the way it is used. Extreme caution is essential in patients who may be hypovolemic; whenever possible, give fluid before anesthetic induction. In the hypovolemic patient, ketamine is less likely than other induction drugs to cause profound hypotension

Continue

- Paralysis by using muscle relaxant agent with short onset of action (≤ 1min)
- Application of cricoid pressure—using one or two hands
- Direct laryngoscopy and oral intubation

Placing the patient's head and neck in neutral alignment will tend to make the view of larynx at laryngoscopy difficult use of a Gum Elastic Bougie and a video-laryngoscope is recommended

If intubation is impossible, a supraglottic airway (e.g. ProSeal LMA or i-Gel) will provide a temporary airway but may not prevent aspiration



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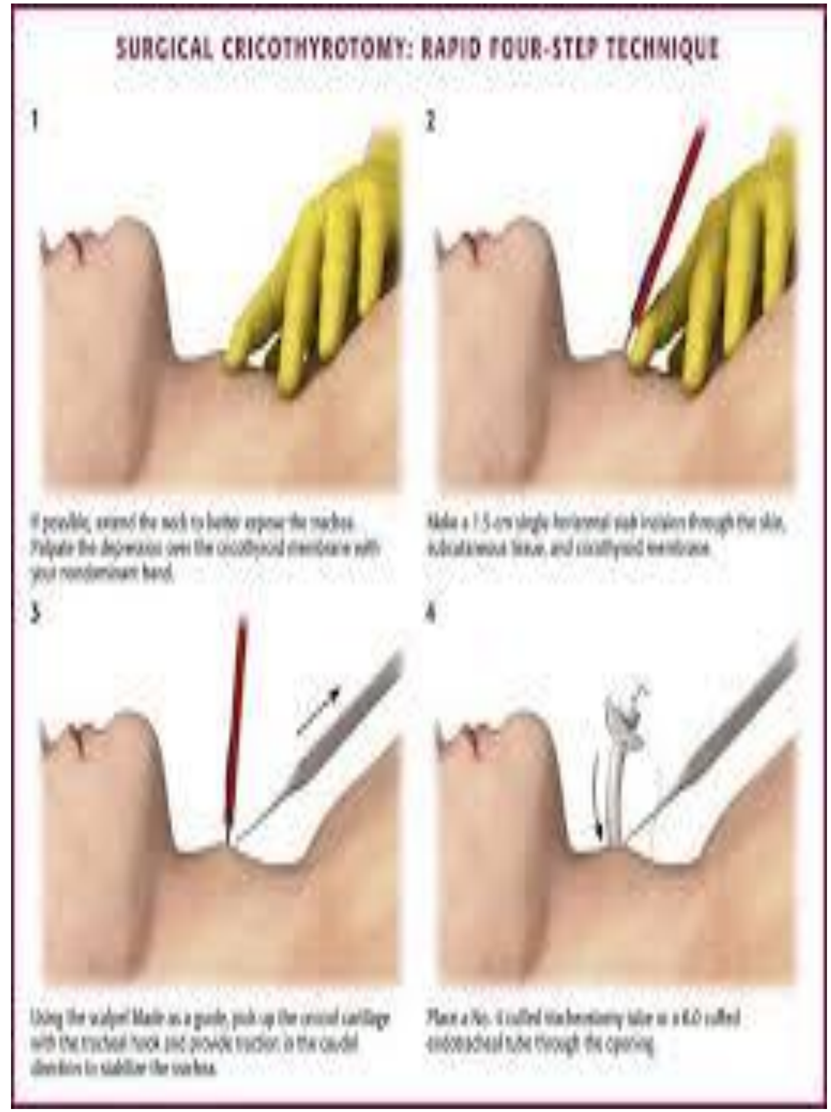
Surgical crico-thyroidotomy, using a scalpel and a 6.0mm internal diameter ETT, is indicated if the patient cannot be intubated by conventional method



Breathing

- Immediately life-threatening chest injuries:

- Tension pneumothorax: Reduced chest movement, reduced breath sounds, and a resonant percussion note on the affected side, along with respiratory distress, hypotension, and tachycardia, indicate a tension pneumothorax.
- Deviation of the trachea to the opposite side is a late sign, and neck veins may not be distended in the presence of hypovolaemia.

- Treatment is immediate decompression with either a large cannula placed in the 2nd intercostal space in the mid-clavicular line on the affected side or a rapid thoracostomy (small incision into the pleural space) in the 5th intercostal space in the anterior axillary line.
- Once IV access has been obtained, insert a large chest drain (32FG) in the 5th intercostal space (anterior axillary line), and connect to an underwater seal drain.

- Open pneumothorax: Cover an open pneumothorax with an occlusive dressing, and seal on three sides; the unsealed side should act as a flutter valve. Insert a chest drain away from the wound in the same hemithorax

-Massive hemothorax

- (defined as >1500mL of blood in a hemithorax) will cause reduced chest movement and a dull percussion note, in the presence of hypoxaemia and hypovolaemia.
- Start fluid resuscitation, and insert a chest drain.
- The patient is likely to require a thoracotomy if blood loss from the chest drain exceeds 200mL per hour, but this decision will depend also on the patient's general physiological state.

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Cardiac Tamponade

- Consider cardiac tamponade while examining the chest, particularly if the patient has sustained a penetrating injury to the chest or upper abdomen.
- Distended neck veins in the presence of hypotension are suggestive of cardiac tamponade, although, after rapid volume resuscitation, myocardial contusion may also present in this way. Tension pneumothorax may also mimic tamponade.
- Muffled heart sounds are meaningless in the midst of a busy resuscitation room.

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- Ultrasound examination in the resuscitation room is the best way to make the diagnosis, focused assessment sonogram in trauma (FAST) scanning in emergency departments.
- If cardiac tamponade is diagnosed or suspected after a penetrating injury and the patient is deteriorating, despite all resuscitative efforts, an urgent thoracotomy and pericardiotomy will be required. This is best undertaken in an operating theatre
- In the absence of a suitably experienced surgeon, pericardiocentesis may provide temporary relief from the tamponade,

Circulation—management of hypovolemia

- Control external haemorrhage with direct pressure.
- Hypovolemic shock is divided into four classes, according to the percentage of total blood volume lost and the associated symptoms and signs.
- Hemorrhage alone, in the absence of significant tissue injury, causes relatively less tachycardia and is easily overlooked, particularly in young, fit people who are able to compensate to a remarkable degree.

- A decrease in systolic pressure suggests a loss of >30% of total blood volume (~1500mL in a 70kg adult).
- Reduced consciousness caused by hypovolaemia implies at least 40–50% loss of blood volume.
- Insert two short, large-bore IV cannulae (14G); take blood samples for FBC and electrolytes, and cross-match from the 1st cannula.
- If peripheral access is difficult, use an external jugular vein or femoral vein (avoid in abdominal/pelvic/leg injury).

- The intraosseous route (usually via the proximal tibia or proximal humerus) is useful route for adults & children. It enable the infusion of fluids at 200mL/min.

- Insert an arterial cannula for continuous direct BP monitoring, and send a sample for ABG analysis—severely injured patients will have a marked base deficit, and its correction will help to confirm adequate resuscitation.

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Classification of hypovolemic shock, according to blood loss (adult)

	Class I	Class II	Class III	Class IV
Blood loss (%)	<15	15–30	30–40	>40
Blood loss (mL)	750	800–1500	1500–2000	>2000
Systolic BP	Unchanged	Normal	Reduced	Very low
Diastolic BP	Unchanged	Raised	Reduced	Unrecordable
Pulse (bpm)	Slight tachycardia	100–120	120 (thready)	>120 (very thready)
Capillary refill	Normal	Slow (>2s)	Slow (>2s)	Undetectable
Respiratory rate	Normal	Tachypnoea	Tachypnoea (>20/min)	Tachypnoea (>20/min)
Urine output (mL/hr)	>30	20–30	10–20	0–10
Extremities	Normal	Pale	Pale	Pale, cold, clammy
Complexion	Normal	Pale	Pale	Ashen
Mental state	Alert	Anxious or aggressive	Anxious, aggressive, or drowsy	Drowsy, confused, or unconscious

Fluids

- Aggressive fluid resuscitation before surgical control of the bleeding is harmful; in the presence of active bleeding, increasing the BP with fluid accelerates the loss of red blood cells and may hamper clotting mechanisms.

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- Untreated hypovolemic shock is associated with microvascular hypoperfusion and hypoxia, leading to multiple organ failure.
- The balance is between the risk of inducing organ ischemia and the risk of accelerating haemorrhage; until haemorrhage control is achieved, the current recommendation is to give 250mL boluses of crystalloid to maintain systolic BP of 80mmHg.

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- Older patients and those with a significant head injury will require a higher BP.

- In an attempt to avoid high volumes of crystalloid in hypovolemic trauma patients, blood and blood products are given much earlier.

Increased survival rates have documented associated with earlier use of platelets and FFP, particularly when given with red cells in ratios approximating 1:1:1.

More recently, it has been suggested that a red cells : FFP ratio of 2:1–1.5:1 may be optimal.

- Tranexamic acid Cyklokapron(1g over 10min, given within 3hr of injury, and then an infusion of 1g over 8hr) reduces mortality from bleeding in trauma patients.

- The use of recombinant factor VII Novo7 may be considered if coagulopathy persists, despite adequate treatment with other blood products.

- A full cross-match will take 45min; group-specific red cells can be issued in 10min, and group O red cells can be obtained immediately

- It is nearly always possible to wait for at least group-specific red cells; major incompatibility reactions when using group-specific red cells are extremely rare

- Once haemorrhage control has been achieved, the goals of fluid resuscitation are to optimize O2 delivery, improve microcirculatory perfusion, and reverse tissue acidosis

Fluid infusion should be targeted at a BP and cardiac output that result in an acceptable urine output and a falling lactate and base deficit

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Warm all IV fluids; a high-capacity fluid warmer is necessary to cope with the rapid infusion rates used during resuscitation of trauma patients

Hypothermia

Hypothermia(core temperature <35°C) is a serious complication and is an independent predictor of mortality.

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Hypothermia has several adverse effects:

- It causes a gradual decline in HR and cardiac output, while increasing the propensity for myocardial dysrhythmias and other morbid myocardial events
- Impairing peripheral O2 delivery in the hypovolemic patient at a time when it is most needed
- Shivering may increase O2 consumption and compound the lactic acidosis that typically accompanies hypovolemia
- Even mild hypothermia inhibits coagulation significantly and increases the incidence of wound infection

Goals for resuscitation of the trauma patient before hemorrhage has been controlled

Parameter	Goal
BP	Systolic 80mmHg. Mean 50–60mmHg
HR	<120bpm
Oxygenation	SaO_2 >95% (peripheral perfusion allowing oximeter to work)
Urine output	>0.5mL/kg/hr
Mental state	Following commands accurately
Lactate level	<1.6mmol/L
Base deficit	>–5
Hb	>8.0g/dL

Disability—rapid neurological assessment

Check the pupils for size and reaction to light, and assess the GCS score rapidly.

If the patient requires urgent induction of anaesthesia and intubation, remember to perform a quick neurological assessment first.

Exposure/environmental control

Undress the patient completely, and protect from hypothermia with warm blankets

Tubes

- Insert a urinary catheter; urine output is an excellent indicator of the adequacy of resuscitation
- Place a gastric tube; this will enable stomach contents to be drained and reduce the risk of aspiration
- If there is any suspicion of a basal skull fracture, use the orogastric route

Imaging

- Whole-body CT as the 1° radiological investigation in severely injured patients
- The standard is to obtain a CT scan within 30min of patient arrival in the emergency department
- Chest and pelvic X-rays are considered if the patient is going directly to the operating room or they are hemodynamically very unstable (e.g. systolic BP < 90mmHg or HR>120bpm despite 2Units of group O-negative blood)
- Any X-rays must be taken without interrupting the resuscitation process—this is achievable if members of the trauma team are wearing lead coats
- A cervical spine injury is assumed until ruled out with a CT scan, with or without clinical examination.

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The secondary survey

- A detailed head-to-toe survey of the trauma patient is undertaken when the vital signs are relatively stable
- Re-evaluate the patient repeatedly, so that ongoing bleeding is detected early
- Patients with exsanguinating haemorrhage may need a laparotomy as part of the resuscitation phase

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Summary

Phases of Management of critically ill Patient includes:

- Primary Survey
- Resuscitation
- Secondary Survey
- Definitive Care
- Tertiary Survey

- Primary Survey

A. Airway
B. Breathing
C. Circulation
D. Disability
E. Exposure

Reference

- Oxford Handbook of Anaesthesia

Questions